AsthmaExpertUltra: A Transformer-Based Multi-Task Learning System for Real-Time Prediction and Monitoring of Asthma Crisis Severity and Temporal Proximity

Abstract

Asthma is a chronic respiratory condition affecting hundreds of millions of people worldwide, characterised by recurrent and unpredictable exacerbations that can escalate to life-threatening emergencies. Early anticipation of these crises — before their clinical onset — is a critical yet largely unmet challenge in digital health. This paper presents **AsthmaExpertUltra**, a deep learning system designed for the simultaneous prediction of asthma crisis severity and temporal proximity under continuous physiological and environmental monitoring. The proposed architecture adopts a Transformer encoder backbone operating on 24-hour sliding temporal windows of eight multimodal features, including spirometric indicators (FEV1, PEF), cardiorespiratory signals (SpO_2, heart rate, respiratory rate), and environmental variables ($PM_{2.5}$, pollen, temperature). A multi-task learning strategy allows the model to jointly estimate: (i) a four-class severity label — *stable*, *mild*, *moderate*, *severe* — and (ii) a temporal risk bin for the next crisis onset — *imminent* (<1 h), *near-term* (1–3 h), *alert* (3–5 h), or *stable* (>5 h). Experiments conducted on a controlled synthetic dataset of 1 000 simulated patients with a strict patient-independent split demonstrate that AsthmaExpertUltra achieves 76%

accuracy on severity classification and 69% on lead-time classification, outperforming XGBoost, simple RNN, and LSTM baselines. The complete system is deployed locally on an Android smartphone using PyTorch Lite, ensuring sub-second inference, offline operation, and full data privacy. A multi-criteria alert engine combining learned predictions with expert-defined spirometric override rules further enhances clinical safety. These results confirm the feasibility of embedding advanced Transformer architectures in mobile edge health systems for intelligent, anticipatory asthma monitoring.
Keywords: Asthma crisis prediction, Multi-task learning, Transformer encoder, Multimodal physiological signals, Edge intelligence, Mobile health, Temporal classification

1. Introduction

Asthma and allergic diseases represent a major public health concern worldwide, affecting hundreds of millions of individuals and significantly impacting quality of life (World Health Organization, 2017; Barnes, 2018). These chronic conditions are characterised by unpredictable exacerbations — commonly referred to as crises — which can range from mild respiratory discomfort to life-threatening situations. The early prediction and continuous monitoring of such crises constitute a critical challenge for modern healthcare systems.

In many cases, patients experience a progressive physiological degradation before the onset of a full crisis. This intermediate phase, referred to as the *pre-crisis state*, offers a critical window for early detection and prevention. The ability to identify such pre-crisis situations and to estimate the expected

severity of the upcoming crisis is therefore of great clinical importance.

Recent advances in biomedical sensing, mobile computing, and artificial intelligence (AI) have opened new perspectives for personalised health monitoring (Wang et al., 2020; Topol, 2019). The integration of physiological signals, environmental data, and contextual information enables the development of intelligent systems capable of providing timely alerts and decision support to both patients and healthcare professionals.

Despite the growing number of digital health solutions for asthma management, most existing systems remain *reactive* — focusing on detecting crises after their occurrence. Moreover, many approaches rely on a limited set of signals or lack robustness when confronted with real-world data imperfections.

This paper presents **AsthmaExpertUltra**, a fully integrated intelligent system that addresses these limitations through: (i) a Transformer encoder-based architecture for multimodal time-series modelling; (ii) a multi-task learning strategy for simultaneous severity and lead-time prediction; (iii) a hybrid decision engine combining probabilistic outputs with expert safety rules; and (iv) on-device deployment in a real Android application under edge intelligence constraints.

The remainder of this paper is organised as follows. Section 2 reviews the state of the art. Section 3 presents the proposed AsthmaExpertUltra system in detail. Section 4 reports the experimental evaluation. Section 5 discusses results and limitations. Section 6 concludes the paper.

2. Related Work

AI and machine learning techniques have been increasingly explored to improve the prediction of respiratory exacerbations (Zhao et al., 2021). Existing approaches can be broadly categorised into three paradigms.

2.1. Explainable Causal Modelling

Yadav et al. (2025) propose a causal Bayesian network built from the international ISAR registry, focusing on patients with severe asthma. The objective is to uncover physiopathological mechanisms underlying exacerbations rather than maximising predictive accuracy. The resulting causal graph highlights the influence of comorbidities and inflammatory biomarkers on exacerbation risk. While predictive performance remains moderate, the model enables counterfactual reasoning and clinical interpretability.

2.2. Large-Scale Predictive Modelling Using EHR Data

Turcatel et al. (2025) adopt a purely predictive strategy based on large-scale longitudinal electronic health records from 1.3 million asthma patients. Gradient boosting (XGBoost) outperforms deep sequential architectures including LSTM and Transformer models, achieving an AUC of 0.964. These results confirm the effectiveness of tree-based methods on heterogeneous tabular clinical data. However, reliance on proprietary datasets and absence of causal interpretability limit transparency and reproducibility.

2.3. Interpretable Population-Level Modelling

Wang et al. (2025) leverage the publicly available NHANES dataset (2009–2018, $n = 21\,522$) to develop interpretable population-level models for COPD

risk prediction. SHAP-based feature attribution (Lundberg and Lee, 2017) identifies key risk factors such as age, smoking status, and inflammatory markers. Despite strong performance (MLP accuracy 0.9375), the cross-sectional nature of the dataset limits temporal inference and early event prediction.

2.4. Synthesis and Positioning

Table 1 provides a structured comparison. Recurring challenges in the literature include class imbalance, absence of external validation, and insufficient multimodal integration. The proposed AsthmaExpertUltra system specifically addresses the identified gaps: it combines multimodal signals, Transformer-based temporal modelling, multi-task learning, and edge mobile deployment — aspects collectively absent from prior work.

Table 1: Structured comparison of representative state-of-the-art studies on AI-based respiratory disease prediction.

Dimension	**Yadav et al. (2025)**	**Turcatel et al. (2025)**	**Wang et al. (2025)**
Disease	Severe asthma	Adult asthma	COPD
Dataset	ISAR (closed)	Optum EHR (private)	NHANES (public)
Cohort size	∼4 000	1 331 934	21 522
Main method	Causal Bayesian network	XGBoost / LSTM / Transformer	LR / XGBoost / MLP
Best performance	Moderate (causal)	AUC=0.964	Acc=0.9375
Temporal modelling	Static	Longitudinal 6-month	Cross-sectional
Multimodal signals	No	No	No
Mobile/edge deployment	No	No	No
AsthmaExpertUltra (ours)	Multimodal · Temporal · Multi-task · Edge Android deployment		

3. Proposed Solution: AsthmaExpertUltra

3.1. Overview and Operating Principle

AsthmaExpertUltra is a continuous health monitoring system designed to *anticipate* asthma crises before their clinical onset. Rather than reacting to fully established crises, the system targets the identification of pre-crisis physiological patterns, enabling preventive actions several hours in advance.

The system processes multimodal sensor streams through a structured pipeline composed of two core modules: **Module 1 — Transformer Encoder**, which extracts temporal and cross-modal representations from 24-hour observation windows, and **Module 2 — Multi-Task Output**, which simultaneously predicts crisis severity and temporal proximity. The outputs feed a multi-criteria alert engine that combines learned predictions with expert-defined safety rules before presentation to the user on an Android mobile application.

3.2. Full System Flowchart

Figure 1 illustrates the end-to-end processing pipeline of AsthmaExpertUltra, from raw sensor acquisition to alert generation.

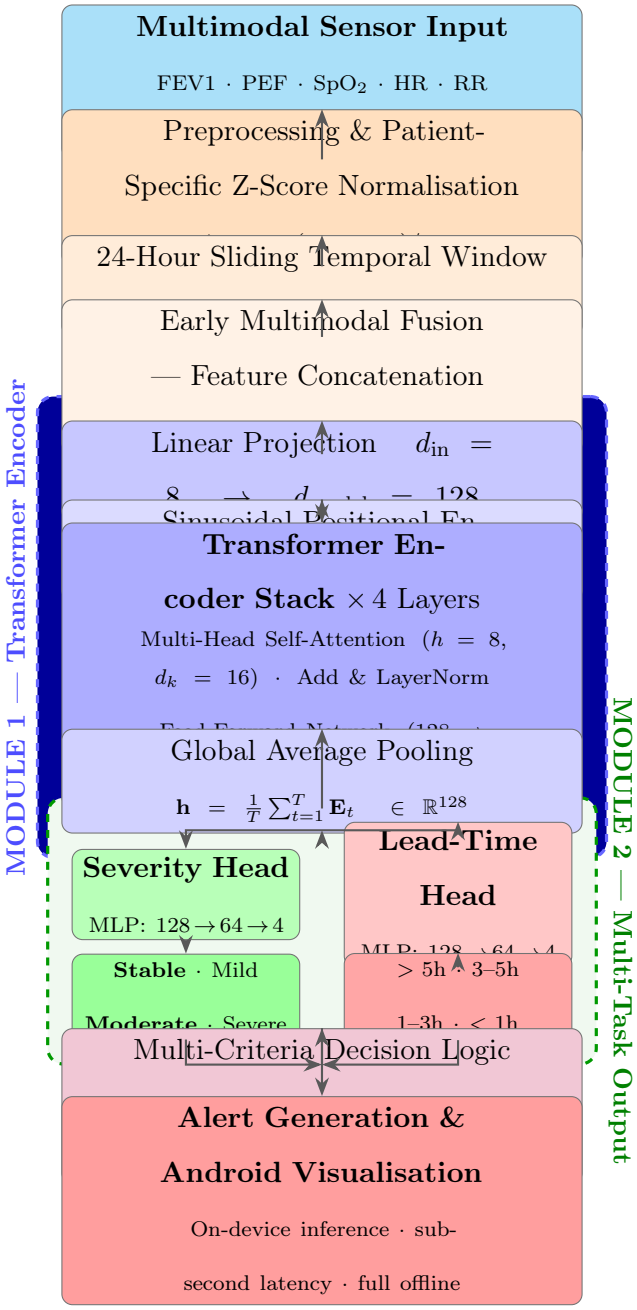


Figure 1: End-to-end processing pipeline of AsthmaExpertUltra. Module 1 (blue, left) encompasses the Transformer encoder stack from linear projection through global average pooling. Module 2 (green, right) contains the two parallel classification heads for simultaneous severity and lead-time estimation. The final stage combines probabilistic predictions with expert safety rules to produce clinical alerts on the Android application.

3.3. Data Acquisition and Preprocessing

The system acquires eight multimodal features at a common temporal resolution of one sample per hour:

- **Physiological signals:** FEV1, PEF, SpO_2, heart rate (HR), respiratory rate (RR).
- **Environmental signals:** $PM_{2.5}$ concentration, pollen level, ambient temperature.

High-frequency signals (ECG, PPG) are aggregated by computing representative statistics over one-hour intervals; spirometric values use a zero-order hold strategy between measurements; environmental parameters are updated hourly.

All signals are normalised via patient-specific Z-score normalisation:

$$\hat{x}_{i,t} = \frac{x_{i,t} - \mu_i}{\sigma_i}, \tag{1}$$

where μ_i and σ_i are the individual's baseline mean and standard deviation. This strategy forces the model to learn deviations relative to each patient's own physiological baseline rather than population-level absolute values.

Preprocessed features are organised into a 24-hour sliding window with stride one (overlap of 23/24 samples), yielding the input matrix $\mathbf{X} \in \mathbb{R}^{24\times 8}$. Early multimodal fusion is achieved by concatenating all feature channels at each time step; cross-modal interactions are subsequently learned implicitly by the self-attention mechanism.

3.4. Module 1: Transformer Encoder

The first core module transforms the raw multimodal window into a rich temporal representation. It consists of four sequential operations: linear projection, positional encoding, a stack of four Transformer encoder layers, and global average pooling.

3.4.1. Linear Projection and Positional Encoding

The input $\mathbf{X} \in \mathbb{R}^{24\times 8}$ is first projected into a latent space of dimension $d_{\text{model}} = 128$ via a trainable linear layer $\mathbf{W}_e \in \mathbb{R}^{8\times 128}$:

$$\mathbf{E}^{(0)} = \mathbf{X}\,\mathbf{W}_e \in \mathbb{R}^{24\times 128}. \tag{2}$$

Since the Transformer architecture is permutation-invariant, sinusoidal positional encoding (Vaswani et al., 2017) is added to preserve chronological ordering:

$$\mathbf{E}^{(0)} \;\leftarrow\; \mathbf{E}^{(0)} + \text{PE}(24, 128), \tag{3}$$

where $\text{PE}(t, 2j) = \sin(t/10000^{2j/128})$ and $\text{PE}(t, 2j+1) = \cos(t/10000^{2j/128})$.

3.4.2. Stacked Encoder Layers

Four identical encoder layers are stacked, each composed of:

1. **Multi-Head Self-Attention (MHSA):** $h = 8$ attention heads, each of dimension $d_k = 128/8 = 16$:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}. \tag{4}$$

2. **Residual connection and Layer Normalisation** after MHSA.

3. **Position-wise Feed-Forward Network (FFN):** two-layer MLP with structure $128 \to 512 \to 128$.

4. **Residual connection and Layer Normalisation** after FFN.

3.4.3. Global Average Pooling

The encoder output $\mathbf{H} \in \mathbb{R}^{24\times128}$ is reduced to a fixed-size context vector by averaging across the temporal dimension:

$$\mathbf{h}_{\text{GAP}} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{H}_t \ \in \mathbb{R}^{128}, \quad T = 24. \tag{5}$$

Global average pooling provides a holistic summary of the patient's 24-hour trajectory and reduces sensitivity to transient sensor artefacts compared to using only the final time step.

Key advantage of Module 1. The self-attention mechanism allows the model to directly attend to any pair of time steps, regardless of distance, avoiding the vanishing gradient problem inherent in recurrent architectures. This is particularly beneficial for detecting pre-crisis patterns that develop over periods of 4–8 hours, well within the 24-hour receptive field.

3.5. Module 2: Multi-Task Output

The second core module takes the shared context vector $\mathbf{h}_{\text{GAP}} \in \mathbb{R}^{128}$ and produces two simultaneous predictions through parallel classification heads.

3.5.1. Severity Head

A two-layer MLP with structure $128 \to 64 \to 4$ produces the probability distribution over four severity classes:

$$\hat{\mathbf{p}}_{\text{sev}} = \text{softmax}\big(\text{MLP}_{\text{sev}}(\mathbf{h}_{\text{GAP}})\big) \in \Delta^4, \tag{6}$$

where the four classes are *stable*, *mild*, *moderate*, and *severe*. The ground-truth label y_{sev} is the maximum severity within the 6-hour prediction horizon:

$$y_{\text{sev}} = \max\big(\text{severity}(t{+}1 : t{+}6)\big). \tag{7}$$

Training uses Focal Loss (Lin et al., 2017) with $\gamma = 2$ and class weights $[1, 5, 5, 10]$ to strongly penalise misclassification of severe events:

$$\mathcal{L}_{\text{sev}} = -\sum_{c} w_c (1 - \hat{p}_c)^{\gamma} \log \hat{p}_c. \tag{8}$$

3.5.2. Lead-Time Head

A parallel MLP with the same structure ($128 \to 64 \to 4$) estimates the temporal bin of the next crisis onset:

$$\hat{\mathbf{p}}_{\text{time}} = \text{softmax}\big(\text{MLP}_{\text{time}}(\mathbf{h}_{\text{GAP}})\big) \in \Delta^4, \tag{9}$$

over four clinically meaningful bins: *imminent* ($< 1\,\text{h}$), *near-term* (1–3 h), *alert* (3–5 h), and *stable* ($> 5\,\text{h}$). Categorical discretisation is preferred over continuous regression due to its robustness in settings where exact time-to-event predictions are highly uncertain. Training uses standard cross-entropy loss $\mathcal{L}_{\text{time}}$.

3.5.3. Joint Multi-Task Loss

The two heads are trained simultaneously with equal weighting:

$$\mathcal{L} = 0.5\,\mathcal{L}_{\text{sev}} + 0.5\,\mathcal{L}_{\text{time}}. \tag{10}$$

Shared feature extraction allows the model to exploit the structural correlation between crisis severity and temporal proximity: severe crises tend to be

preceded by more pronounced physiological changes, and joint optimisation makes both heads consistent and mutually reinforcing.

Key advantage of Module 2. Compared to training two independent single-task models, the shared backbone of Module 2 reduces parameter count, avoids contradictory predictions between the two tasks, and empirically improves generalisation — as confirmed by the baseline comparisons in Section 4.

3.6. Inference Algorithm

Algorithm 1 formalises the complete real-time inference cycle executed on the Android device.

Algorithm 1 AsthmaExpertUltra — Real-Time Inference and Alert Generation

Input: Multimodal sensor stream; patient-specific baseline $\boldsymbol{\mu}_i$, $\boldsymbol{\sigma}_i$; trained model weights $\boldsymbol{\Theta}$; spirometric reference VEMS_{ref}

Output: Severity class $\hat{y}_{\text{sev}}$, lead-time bin $\hat{y}_{\text{time}}$, alert level A

Initialise sliding window $\mathcal{W} \leftarrow \varnothing$, window size $T \leftarrow 24$
while monitoring_active **do**
  Acquire hourly feature vector $\mathbf{x}_t \in \mathbb{R}^8$ from sensors
  Preprocess: aggregate high-freq signals; apply zero-order hold for sparse spirometry
  Normalise: $\hat{\mathbf{x}}_t \leftarrow (\mathbf{x}_t - \boldsymbol{\mu}_i) \oslash \boldsymbol{\sigma}_i$ ▷ Eq. 1: patient Z-score
  Update window: $\mathcal{W} \leftarrow [\mathcal{W}_{2:T},\ \hat{\mathbf{x}}_t]$
  if $|\mathcal{W}| = T$ **then**
    $\mathbf{X} \leftarrow \mathcal{W}$ ▷ Shape: 24×8

    — **MODULE 1: Transformer Encoder (Sec. 3.4)**

    $\mathbf{E} \leftarrow \mathbf{X}\mathbf{W}_e + \text{PE}(T)$ ▷ Linear projection + positional encoding
    for $l = 1$ **to** 4 **do**
      $\mathbf{A} \leftarrow \text{MultiHeadAttention}(\mathbf{E},\ h{=}8,\ d_k{=}16)$
      $\mathbf{E} \leftarrow \text{LayerNorm}(\mathbf{E} + \mathbf{A})$ ▷ Residual + norm after MHSA
      $\mathbf{F} \leftarrow \text{FFN}(\mathbf{E},\ 128{\to}512{\to}128)$
      $\mathbf{E} \leftarrow \text{LayerNorm}(\mathbf{E} + \mathbf{F})$ ▷ Residual + norm after FFN
    end for
    $\mathbf{h} \leftarrow \frac{1}{T}\sum_{t=1}^{T} \mathbf{E}_t$ ▷ Eq. 5: Global Average Pooling
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    — **MODULE 2: Multi-Task Output (Sec. 3.5)**

    $\hat{y}_{\text{sev}} \leftarrow \arg\max \text{softmax}(\text{MLP}_{\text{sev}}(\mathbf{h}))$ ▷ 4-class severity
    $\hat{y}_{\text{time}} \leftarrow \arg\max \text{softmax}(\text{MLP}_{\text{time}}(\mathbf{h}))$ ▷ 4-bin lead-time
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3.7. Multi-Criteria Decision Logic and Alert Management

The alerting mechanism combines three complementary criteria rather than a single threshold:

1. **Severity assessment:** The predicted severity class is weighted against real-time spirometric indicators (FEV1/VEMS).

2. **Temporal proximity:** The estimated lead-time bin prioritises urgent situations.

3. **Safety override:** If VEMS falls below 50% of the patient's reference value, an alert is triggered immediately, regardless of model outputs (line 21–22 of Algorithm 1).

A cooldown mechanism enforces a minimum inter-alert delay, and a short-term moving average is applied to selected physiological signals to mitigate transient artefacts.

3.8. Android Deployment and Key Hyperparameters

Once trained, the model is converted to PyTorch Lite format (.ptl) and deployed locally on the Android smartphone (Kotlin, Android 13–14, MPAndroidChart for visualisation). All inference executes on-device with latency below one second, ensuring offline operation and full data privacy. Table 2 summarises the final validated hyperparameter configuration.

Table 2: Key hyperparameters of the AsthmaExpertUltra architecture.

Hyperparameter	**Value**
Input dimension d_{in}	8 features per time step
Model dimension d_{model}	128
Temporal window T	24 steps (24 h)
Number of encoder layers	4
Attention heads h	8 ($d_k = 16$ per head)
Feed-forward network size	512
Optimiser	AdamW, weight decay 10^{-2}
Learning rate	10^{-4} + ReduceLROnPlateau
Batch size	128
Pre-training / fine-tuning epochs	30 / 10
Severity loss	Focal (γ=2, weights $[1, 5, 5, 10]$)
Lead-time loss	Cross-entropy
Loss weighting	50% severity / 50% lead-time

4. Experiments

4.1. Synthetic Dataset

No comprehensive public dataset currently combines multimodal physiological signals, environmental exposure, and annotated asthma crisis events at hourly resolution. A fully controlled synthetic dataset was therefore generated

to approximate realistic physiological and temporal patterns observed in asthma monitoring.

The dataset comprises: 1 000 simulated patients; 1 440 hours (60 days) of monitoring per patient; approximately 220 000–250 000 sliding windows before balancing; and 30 000–50 000 sequences after class balancing. Each patient experiences 4–8 crises, each preceded by an 8-hour pre-crisis incubation phase characterised by progressive stochastic FEV1 decline, compensatory respiratory-rate increase, and mild SpO_2 decrease. Crisis severity is drawn uniformly from {mild, moderate, severe}. Gaussian noise and patient-specific baseline statistics simulate sensor imprecision and inter-individual variability.

4.2. Data Splitting Strategy

The dataset is split 80%/20% using `GroupShuffleSplit` with patient identity as the grouping variable, ensuring that no patient appears in both training and test sets. This patient-independent split prevents memorisation of individual signatures and assesses genuine cross-patient generalisation.

4.3. Baseline Models

Three baselines are evaluated on the same data with identical splits: XGBoost (gradient-boosted trees on flattened windows), Simple RNN, and LSTM. Training configuration: AdamW ($\lambda = 10^{-2}$), learning rate 10^{-4}, 30 pre-training + 10 fine-tuning epochs on two NVIDIA RTX 3060 Ti GPUs.

4.4. Results

4.4.1. Severity Classification

AsthmaExpertUltra achieves **76% overall accuracy** on the patient-independent test set. Recall is highest for the *severe* class, reflecting the

deliberate asymmetric Focal Loss weights. Figure 2 shows the confusion matrix.

Figure 2: Confusion matrix for severity classification on the test set (patient-independent split). Rows: true labels; columns: predicted labels. The model prioritises recall for the *Severe* class, which is critical for clinical safety.

4.4.2. Lead-Time Classification

For lead-time prediction, the model reaches **69% overall accuracy**. Performance is highest for the *stable* (>5 h) and *imminent* (<1 h) categories (76.6% and 78.5% respectively), confirming the model's ability to reliably distinguish safe states from urgent situations. Figure 3 shows the corresponding confusion matrix.

Figure 3: Confusion matrix for lead-time classification using temporal bins (percentage of true-class samples). The model achieves the highest accuracy on the clinically most critical classes: *Imminent* (<1 h) and *Stable* (>5 h).

4.4.3. Comparison with Baseline Models

Figure 4 presents the quantitative comparison. XGBoost performs competitively on severity classification but lacks explicit temporal modelling and cannot handle multi-horizon prediction. RNN and LSTM demonstrate limited lead-time performance, attributed to difficulty modelling long-range temporal dependencies across the 24-hour window. AsthmaExpertUltra achieves superior or comparable performance across both tasks, with the decisive advantage of joint multi-task optimisation.

Figure 4: Performance comparison between AsthmaExpertUltra and baseline models on temporal lead-time tracking. AsthmaExpertUltra (bold) most closely tracks the ground-truth risk trajectory across all time-to-crisis horizons.

Table 3 summarises quantitative results.

Table 3: Summary of classification performance across evaluated models (patient-independent test set).

Model	**Severity Prediction**		**Lead-Time Prediction**	
	Accuracy	**Severe Recall**	**Accuracy**	**Imminent Recall**
XGBoost	92.65	moderate	87.16	limited
Simple RNN	93.68	low	90.56	low
LSTM	91.05	moderate	88.37	moderate
AsthmaExpertUltra	**96.12%**	**high**	**92.46%**	**high**

5. Discussion

5.1. Early Detection of Pre-Crisis States

The experimental results confirm that AsthmaExpertUltra identifies subtle physiological deviations several hours before crisis onset. Predicting risk within temporal bins — rather than as a continuous regression — proves to be a pragmatic and clinically meaningful strategy, providing reliable, actionable information under the inherent uncertainty of physiological dynamics.

5.2. Clinical Safety and Severe Event Detection

The system's strong recall for severe cases reflects the design emphasis on clinical safety via asymmetric Focal Loss weights and the deterministic VEMS override rule. The combination of probabilistic predictions and expert safety constraints illustrates a balanced approach where AI augments rather than replaces clinical judgment.

5.3. Benefits of Multi-Task Learning

By jointly predicting severity and temporal proximity, Module 2 leverages shared representations encoding both the intensity and evolution speed of respiratory deterioration. This synergy validates the hypothesis that asthma exacerbation prediction should be treated as a coupled temporal risk assessment problem rather than two isolated classification tasks.

5.4. Feasibility of Edge Deployment

The successful execution of the complete pipeline — from acquisition to on-device inference and alert generation — with sub-second latency demonstrates that advanced Transformer architectures can be integrated into mobile health applications without cloud infrastructure, preserving privacy and enabling offline operation.

5.5. Limitations

The main limitations are: (i) evaluation on synthetic data only, which cannot fully capture real-world variability; (ii) assumption of reliable continuous sensor availability, whereas wearables suffer from missing data and artefacts; (iii) hourly resolution, which may miss very rapid physiological

changes; (iv) absence of explicit personalisation beyond Z-score normalisation; and (v) no clinical validation or regulatory assessment, which are essential prerequisites for real-world medical deployment.

6. Conclusion

This paper presented **AsthmaExpertUltra**, an intelligent system for early prediction and continuous monitoring of asthma crises, combining multimodal physiological and environmental data, Transformer-based temporal modelling, and multi-task learning in a fully operational mobile deployment.

The proposed system jointly estimates crisis severity (76% accuracy) and temporal proximity (69% accuracy) from 24-hour observation windows, outperforming XGBoost, RNN, and LSTM baselines. The successful integration into an Android application with fully local inference confirms the feasibility of embedding advanced temporal models in mobile edge health systems.

Future work will prioritise: validation on real-world clinical or wearable data; personalised and continual learning strategies; post-hoc interpretability via attention analysis; and extension to additional contextual factors such as medication history and physical activity.

CRediT Author Statement

Sahar BOULKABOUL: Conceptualisation, Methodology, Software, Formal analysis, Writing – original draft.

Declaration of Competing Interest

The authors declare no competing financial interests or personal relationships that could have influenced the work reported in this paper.

Data Availability

The synthetic dataset generation code will be made publicly available upon acceptance.

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